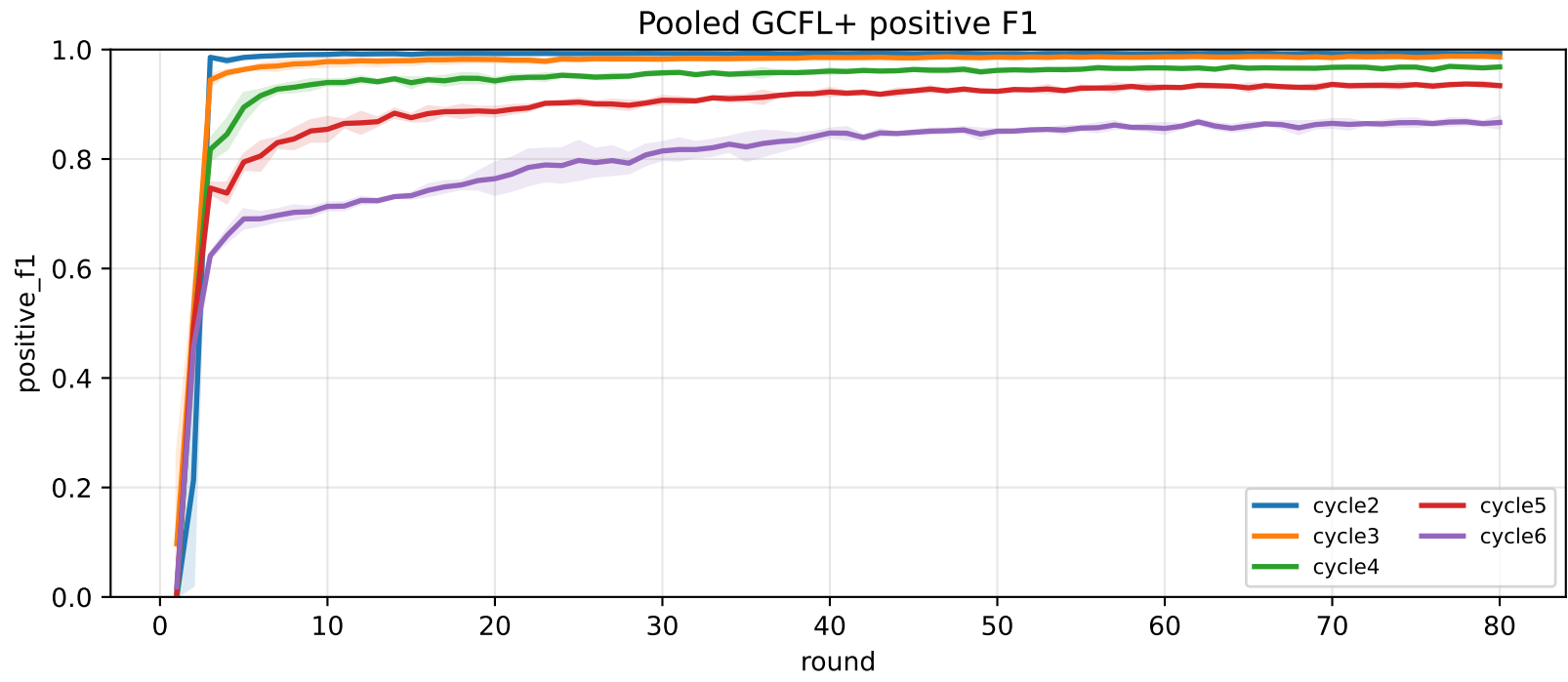
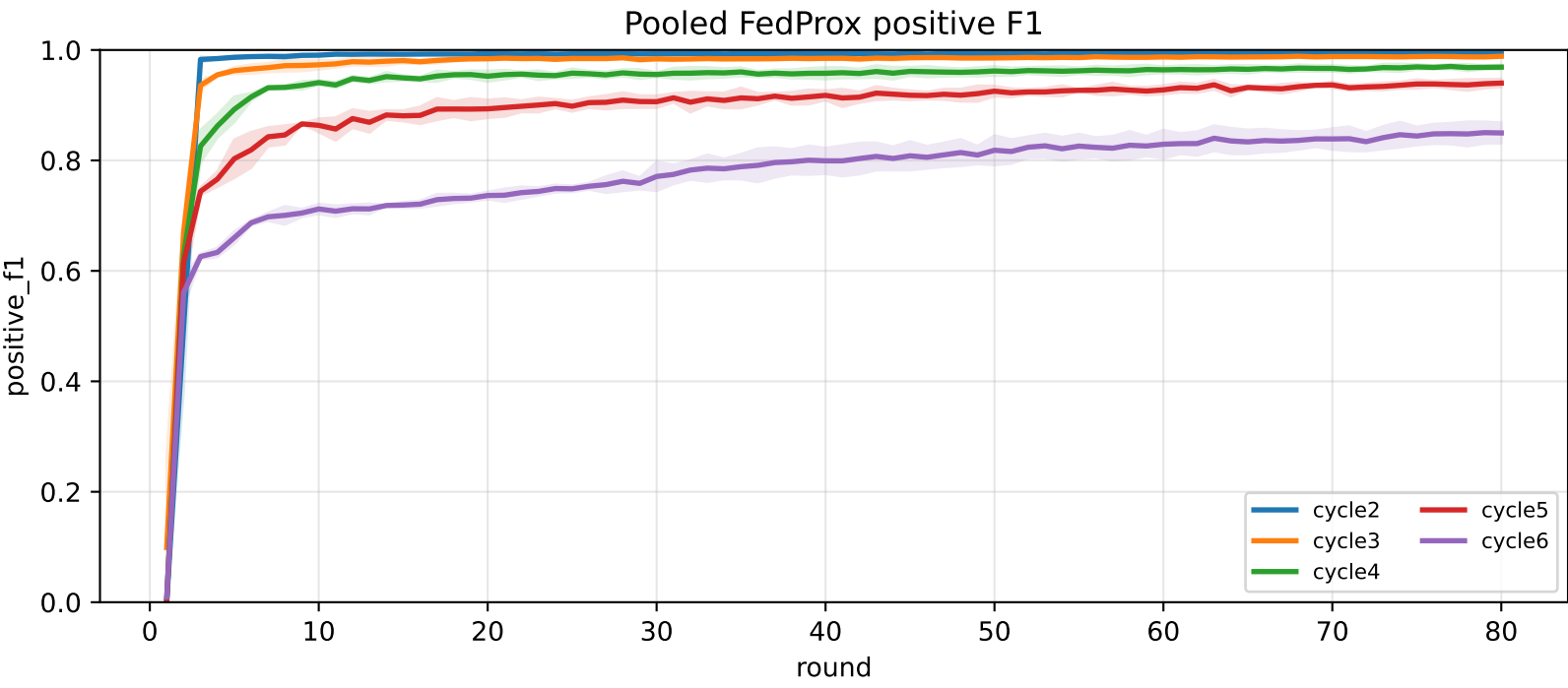
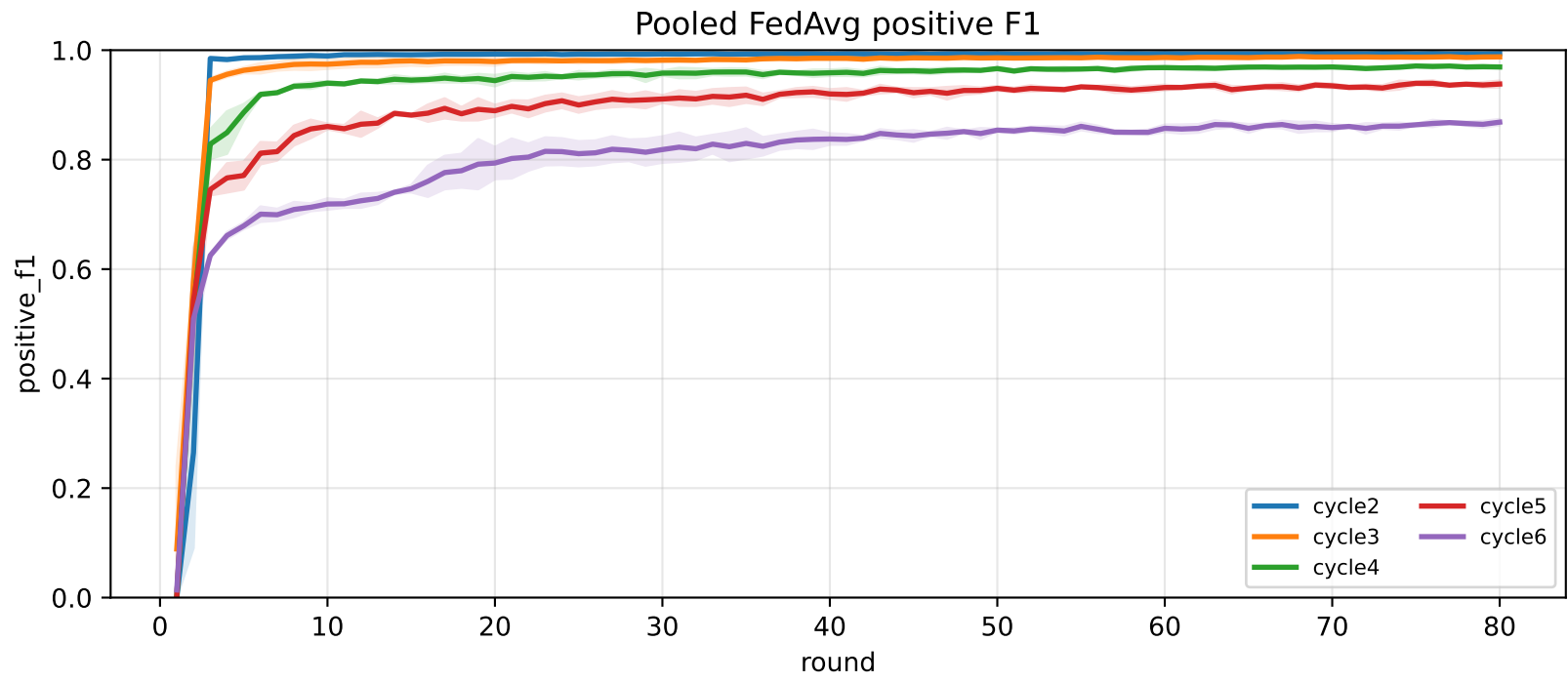
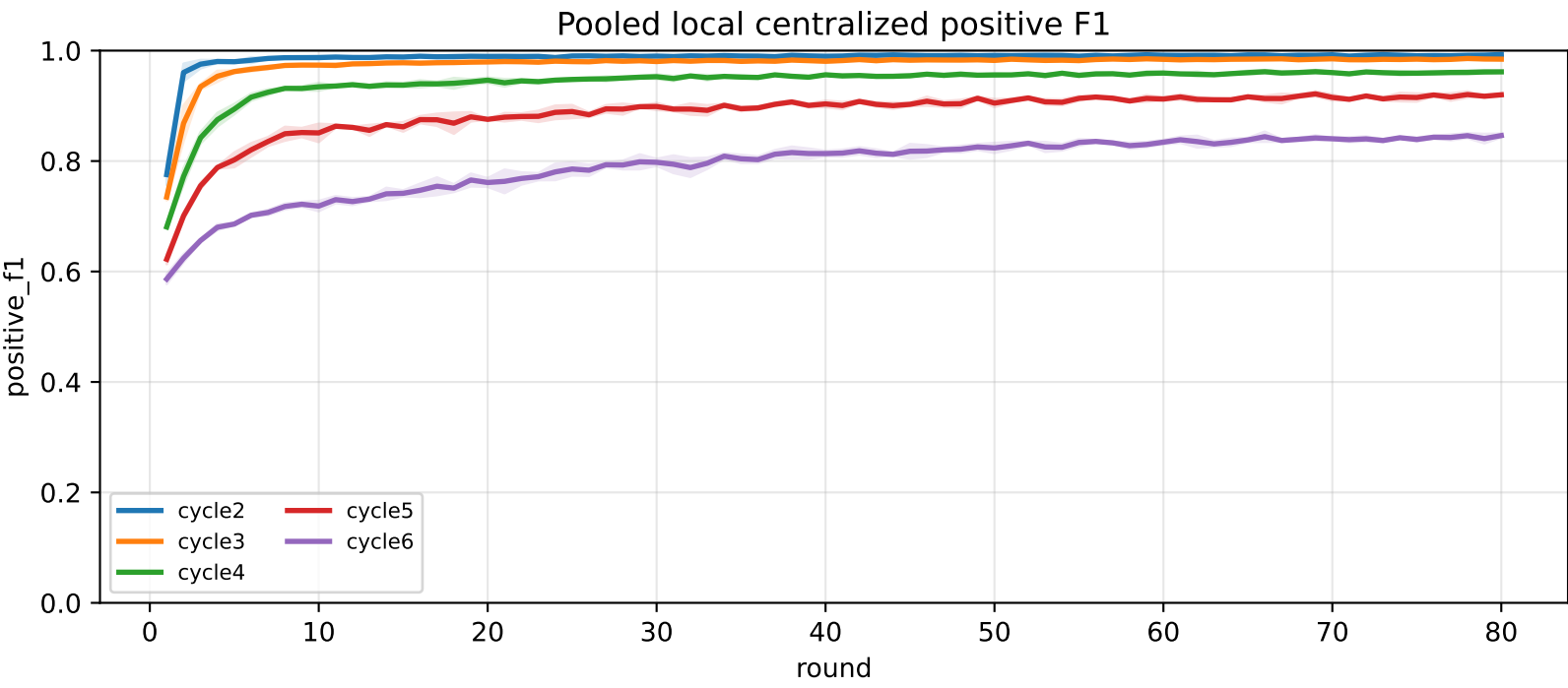
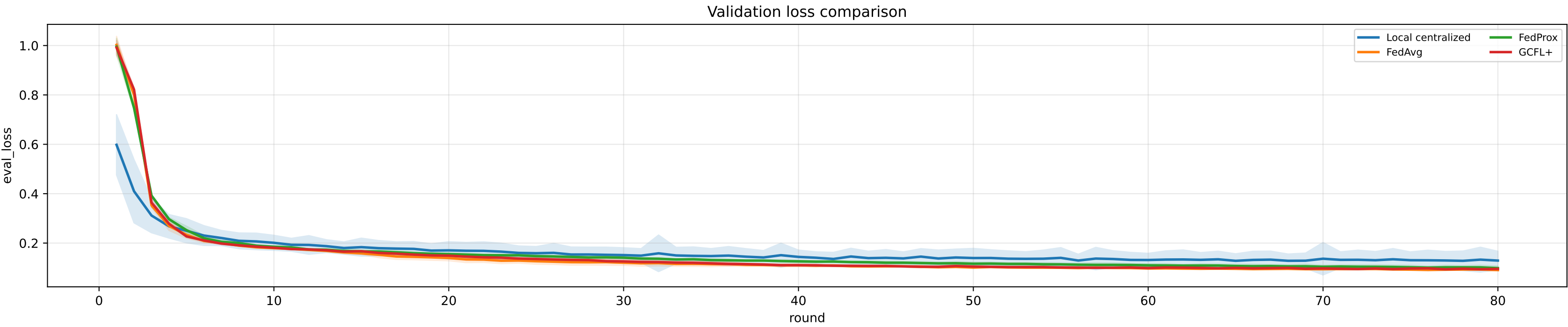


Heterogeneity level: Mid | aggregated / pooled performance over 20 clients |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

Best logged test performance (mean $\pm$ std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.0% $\pm$ 0.1%	93.5% $\pm$ 0.1%	99.2% $\pm$ 0.1%	98.5% $\pm$ 0.0%	96.4% $\pm$ 0.0%	92.1% $\pm$ 0.4%	84.6% $\pm$ 0.3%
FedAvg	95.3% $\pm$ 0.0%	94.9% $\pm$ 0.0%	99.4% $\pm$ 0.0%	98.7% $\pm$ 0.0%	97.2% $\pm$ 0.1%	94.2% $\pm$ 0.2%	87.5% $\pm$ 0.2%
FedProx	94.7% $\pm$ 0.5%	94.3% $\pm$ 0.5%	99.4% $\pm$ 0.0%	98.9% $\pm$ 0.0%	97.1% $\pm$ 0.3%	94.0% $\pm$ 0.6%	85.2% $\pm$ 1.7%
GCFL+	95.0% $\pm$ 0.2%	94.6% $\pm$ 0.2%	99.3% $\pm$ 0.0%	98.7% $\pm$ 0.0%	96.8% $\pm$ 0.2%	93.8% $\pm$ 0.3%	87.2% $\pm$ 0.3%



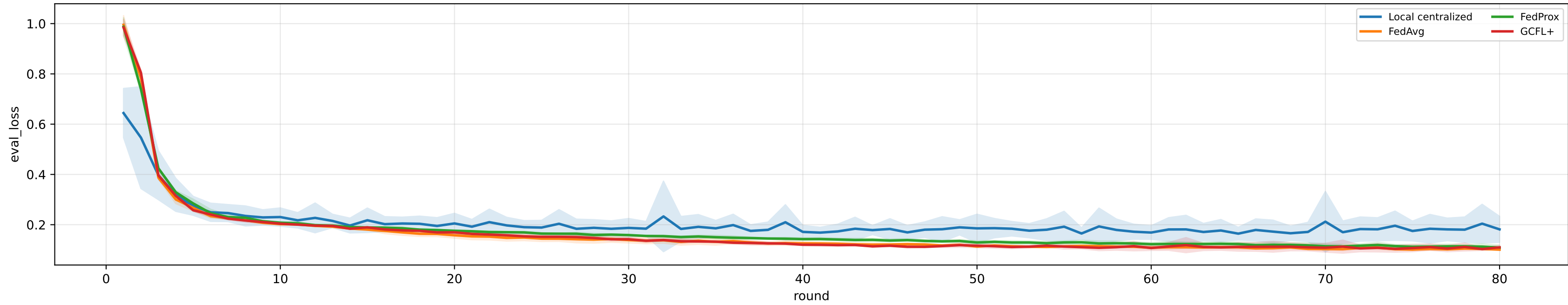
Heterogeneity level: Mid | family aggregated / pooled performance |
family=strong_decreasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_decreasing | cycle2=40.2%±4.0% | cycle3=32.9%±0.8% | cycle4=24.4%±1.8% | cycle5=16.8%±2.6% | cycle6=11.2%±3.3%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

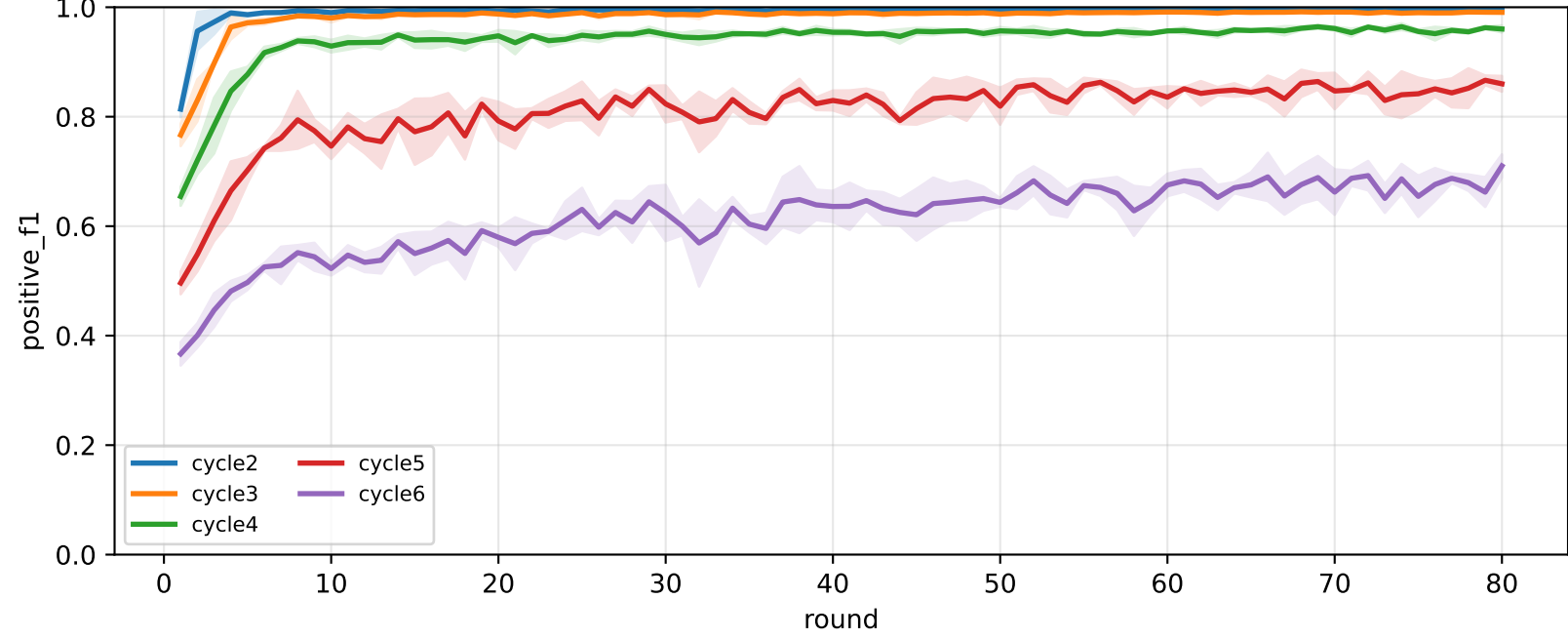
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.1% ± 0.5%	89.3% ± 0.6%	99.6% ± 0.1%	99.0% ± 0.0%	95.8% ± 0.5%	86.3% ± 0.9%	68.1% ± 2.2%
FedAvg	94.9% ± 0.1%	91.7% ± 0.2%	99.8% ± 0.0%	99.4% ± 0.0%	96.8% ± 0.3%	91.0% ± 0.5%	73.8% ± 0.7%
FedProx	94.5% ± 0.5%	91.2% ± 0.7%	99.8% ± 0.0%	99.5% ± 0.0%	97.2% ± 0.2%	90.9% ± 0.9%	70.8% ± 2.6%
GCFL+	94.9% ± 0.3%	91.7% ± 0.4%	99.8% ± 0.1%	99.4% ± 0.0%	96.5% ± 0.6%	91.0% ± 0.7%	74.3% ± 0.6%

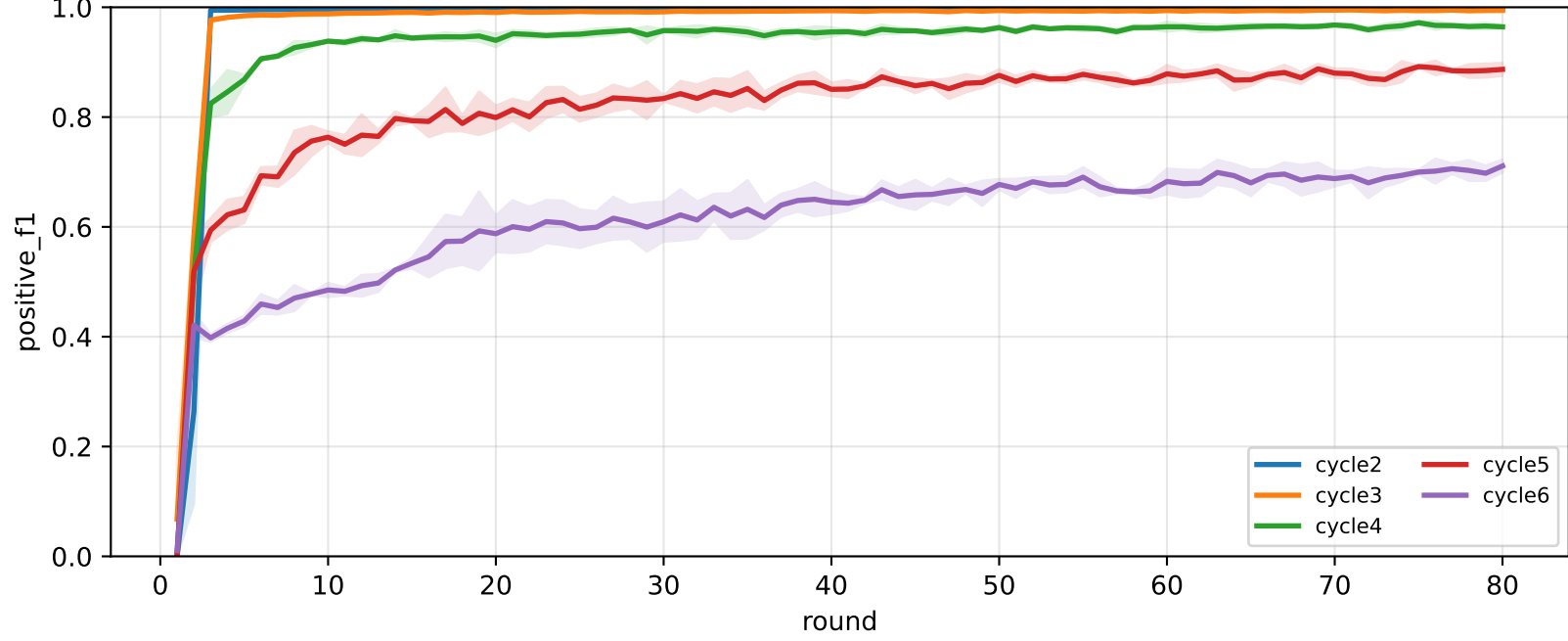
Validation loss comparison



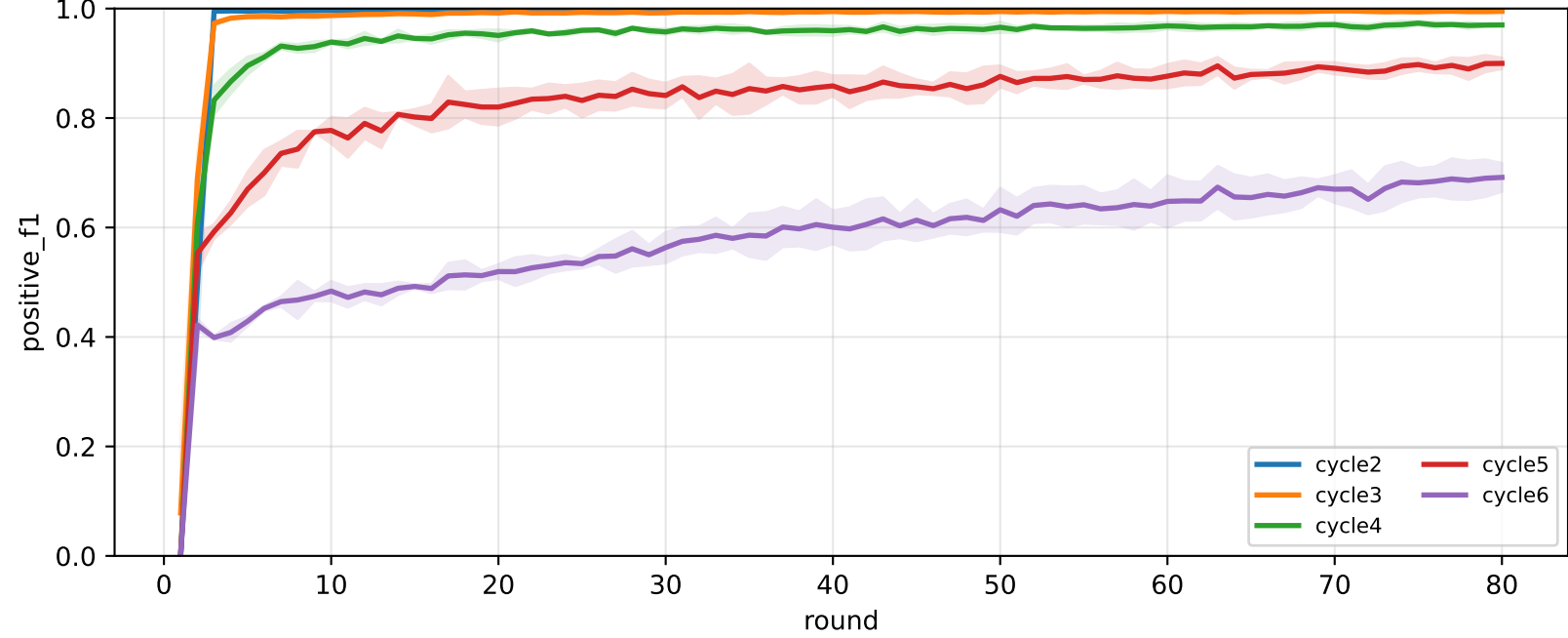
Pooled local centralized positive F1



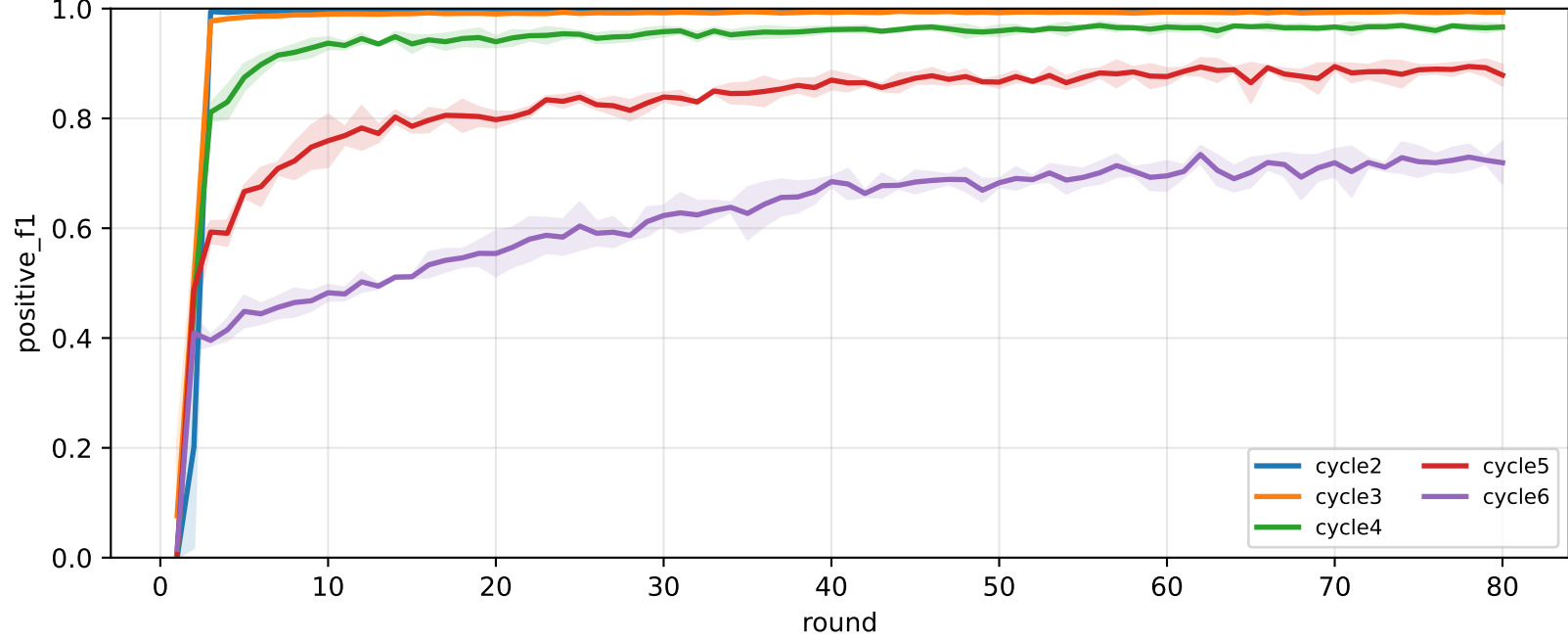
Pooled FedAvg positive F1



Pooled FedProx positive F1



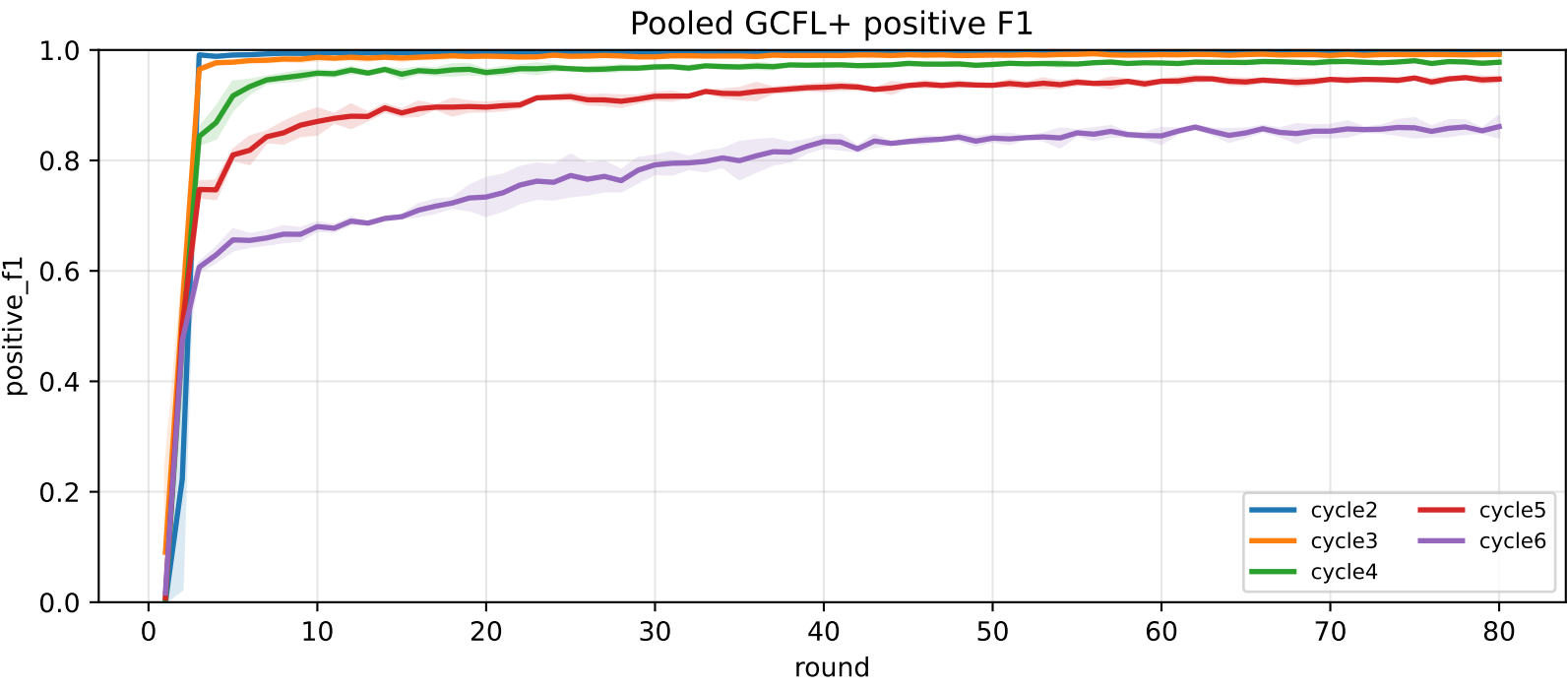
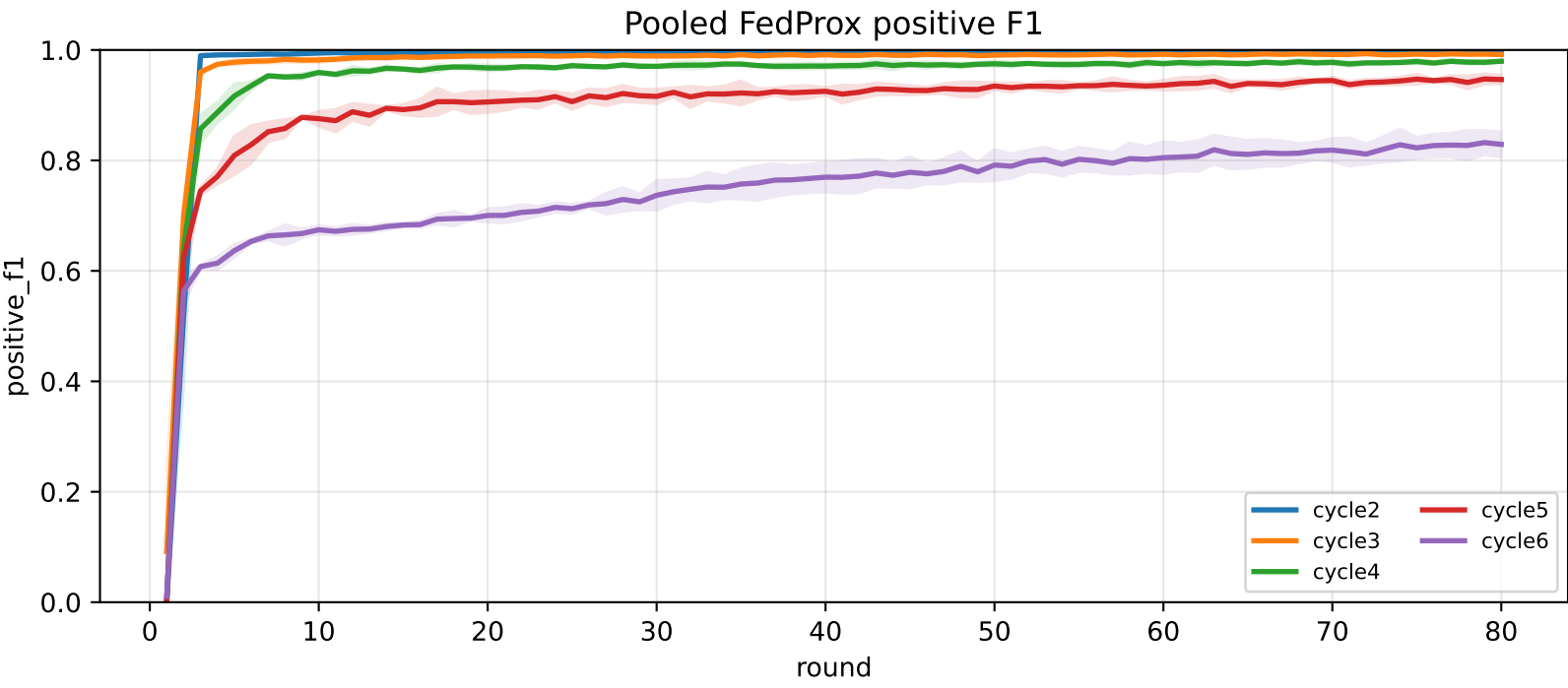
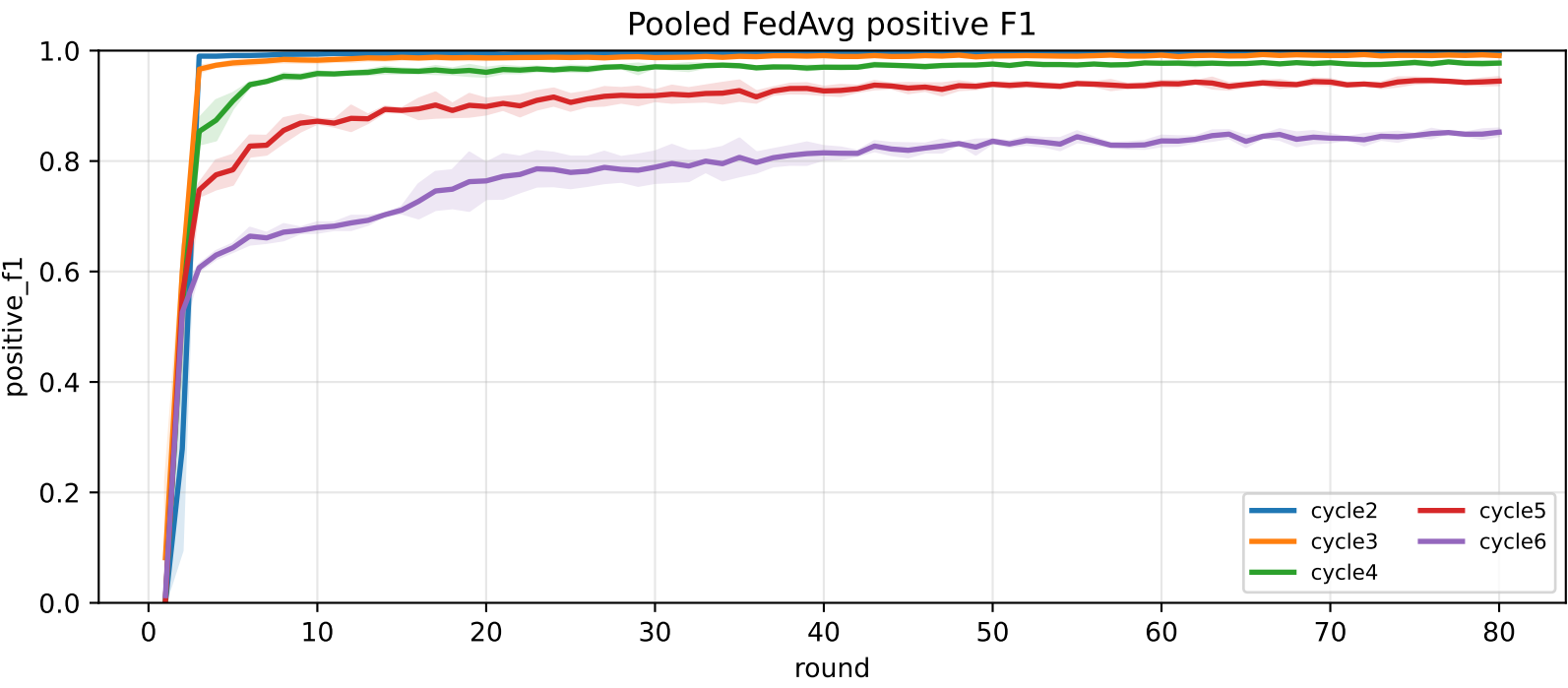
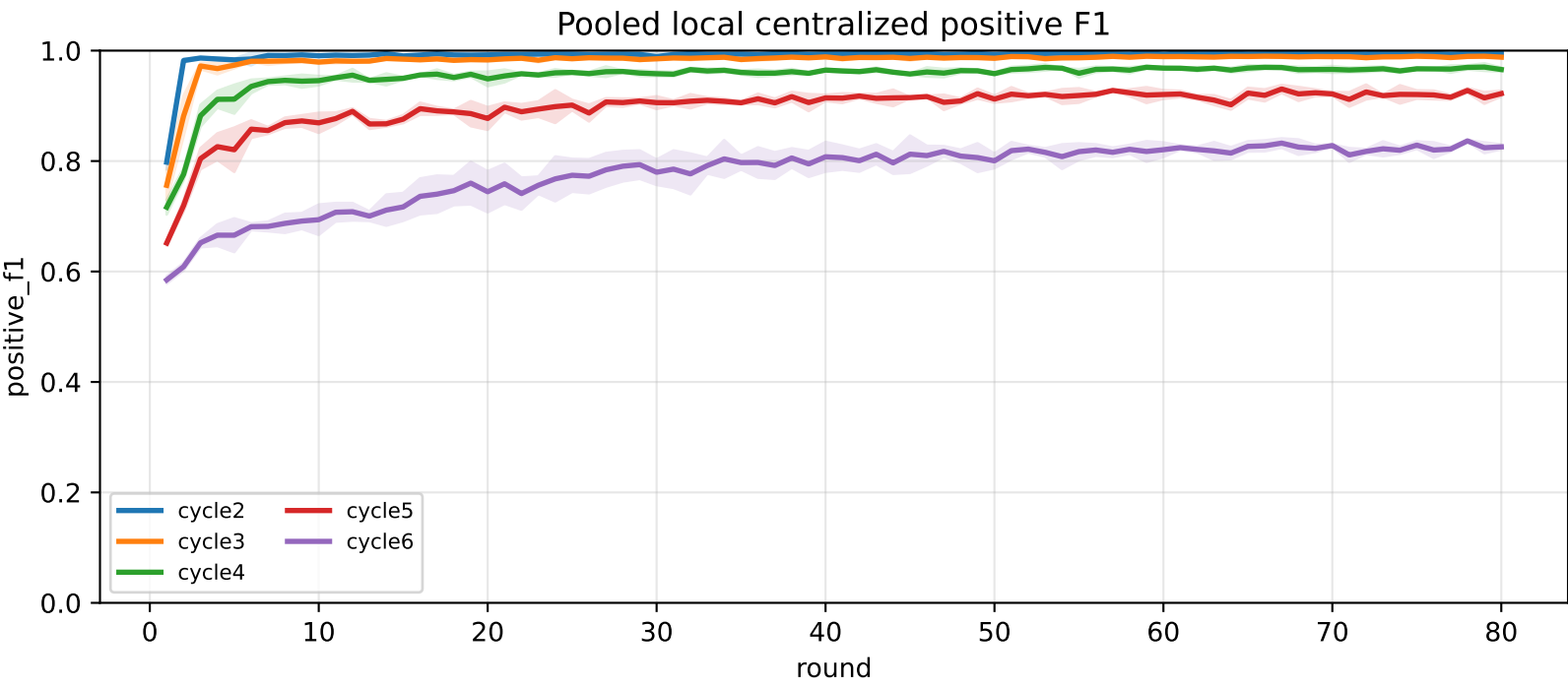
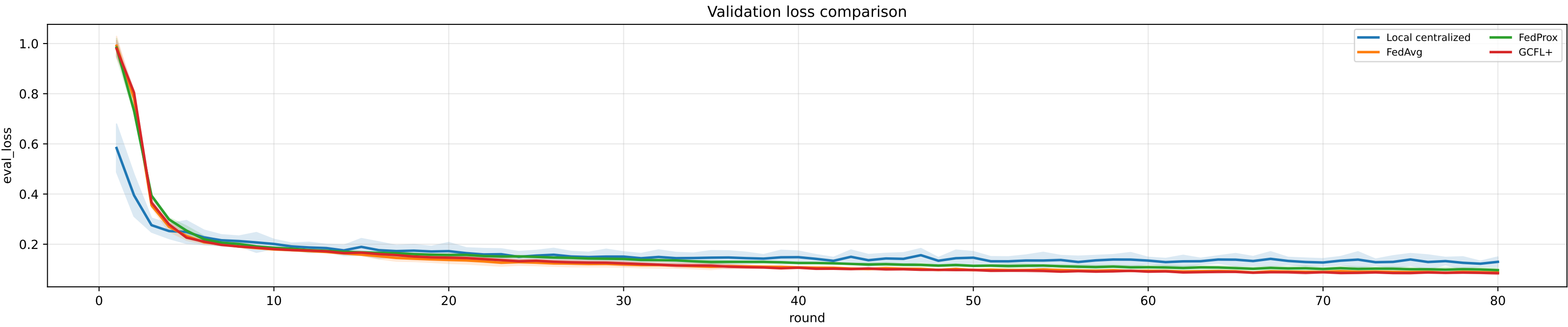
Pooled GCFL+ positive F1



Heterogeneity level: Mid | family aggregated / pooled performance |
family=mild_decreasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_decreasing | cycle2=31.7%±0.3% | cycle3=30.3%±0.5% | cycle4=28.0%±0.8% | cycle5=25.5%±0.5% | cycle6=20.2%±1.0%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

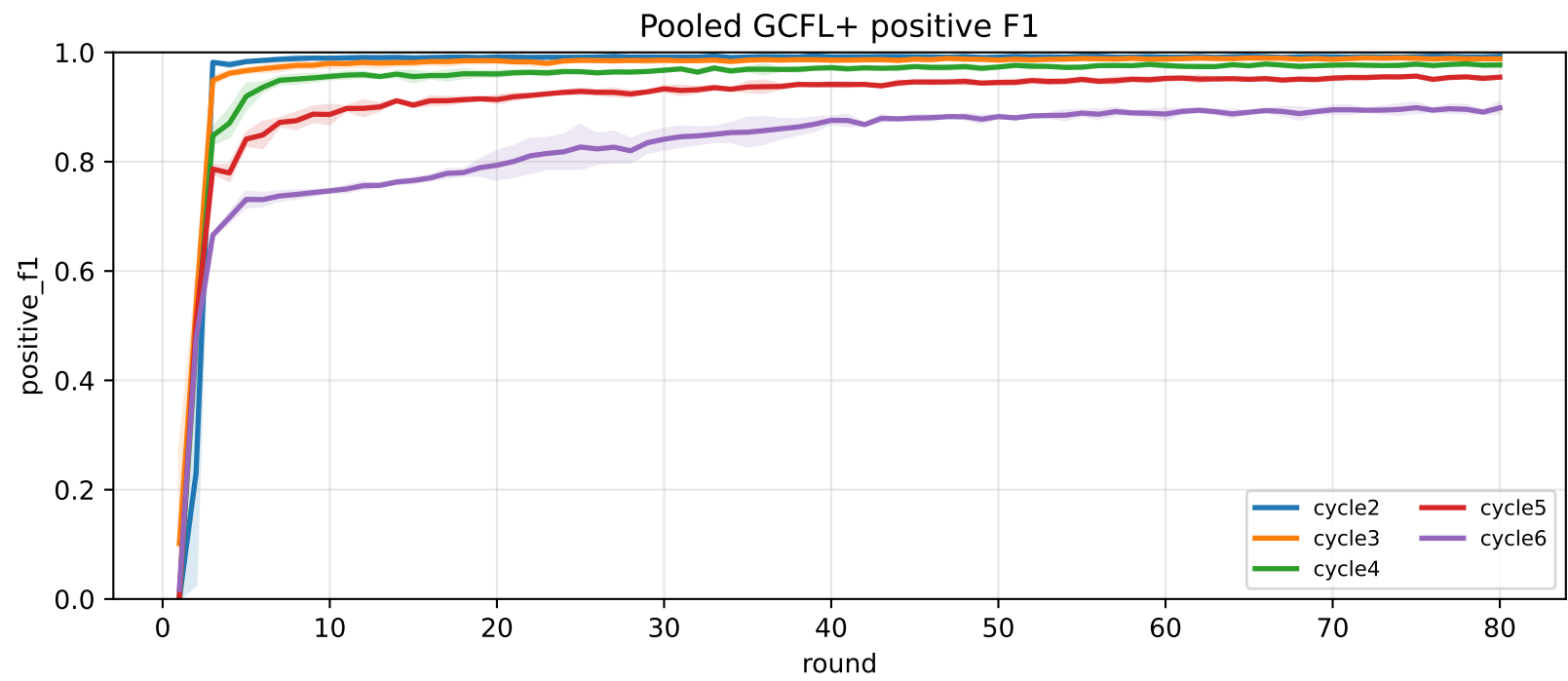
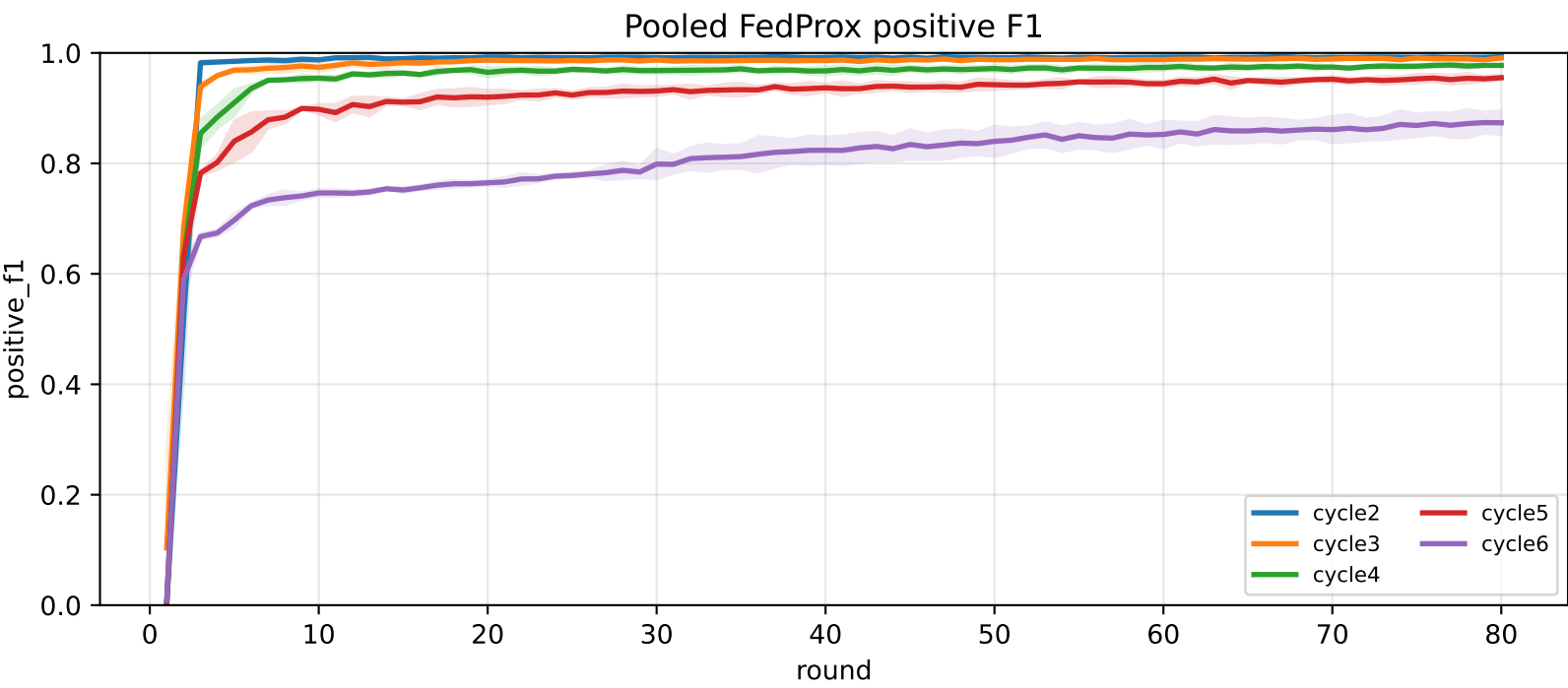
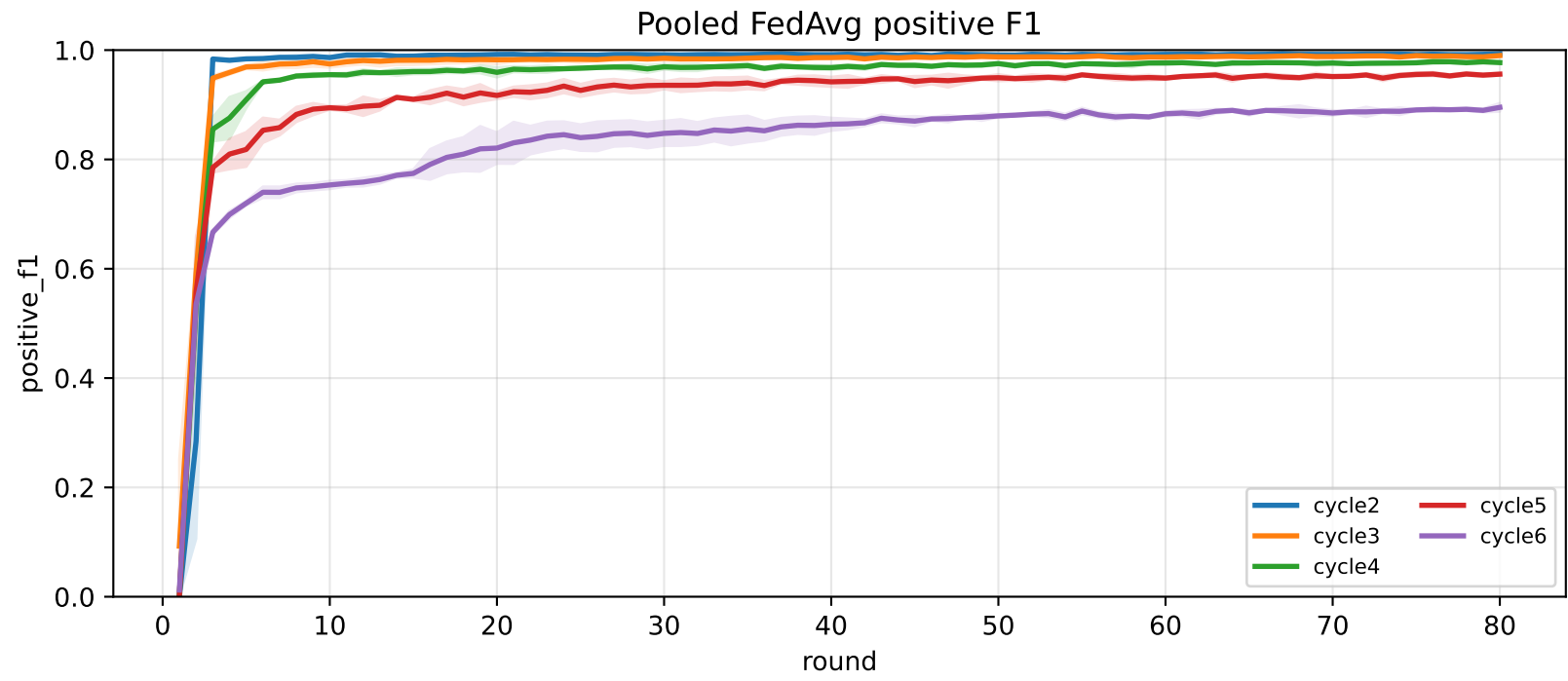
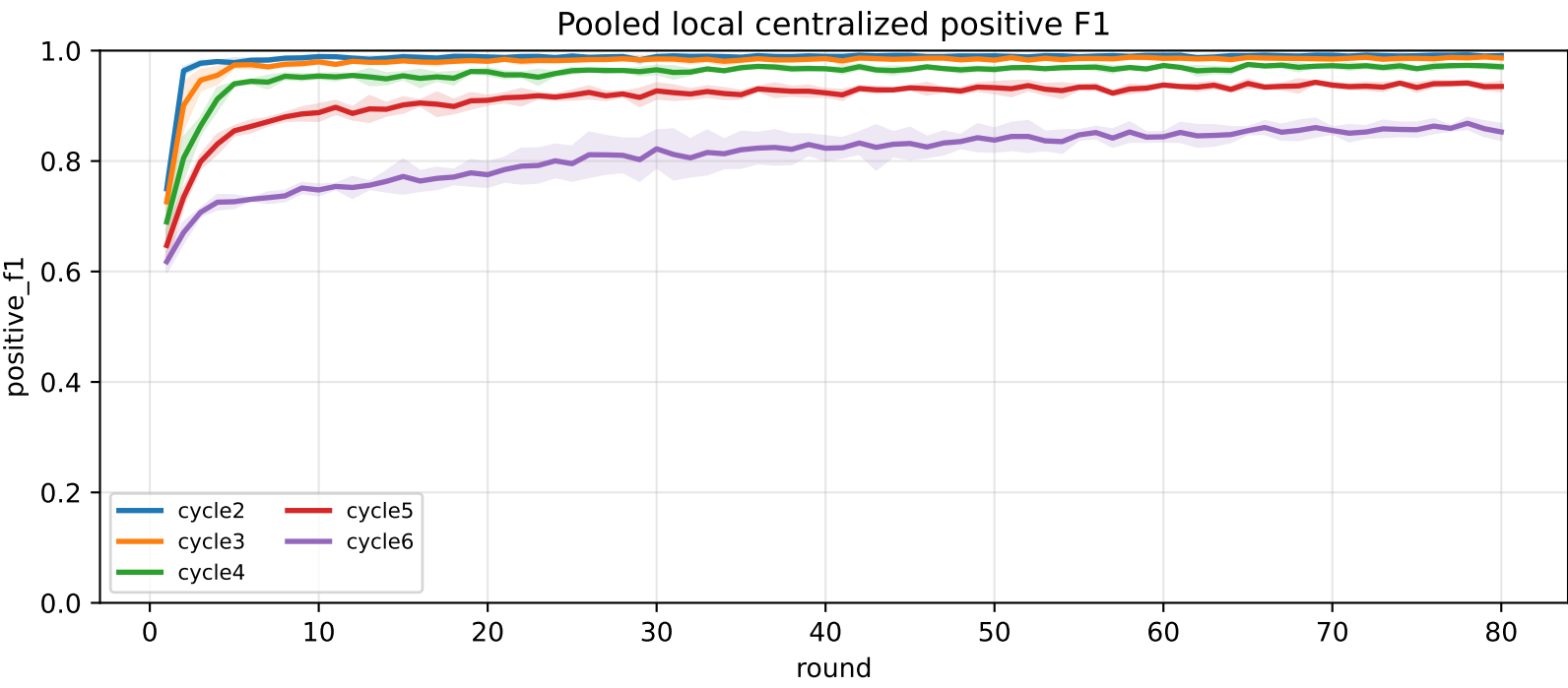
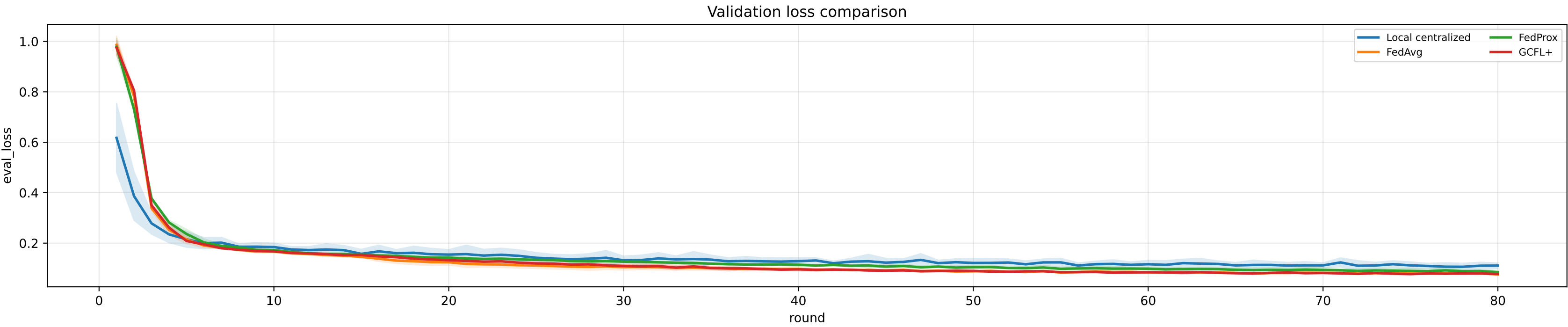
Best logged test performance (mean ± std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.9% ± 0.2%	94.3% ± 0.2%	99.4% ± 0.0%	99.1% ± 0.0%	97.2% ± 0.1%	92.7% ± 0.7%	83.0% ± 0.3%
FedAvg	95.9% ± 0.1%	95.3% ± 0.1%	99.5% ± 0.0%	99.2% ± 0.0%	97.8% ± 0.1%	94.9% ± 0.4%	85.2% ± 0.6%
FedProx	95.3% ± 0.5%	94.7% ± 0.5%	99.5% ± 0.1%	99.3% ± 0.0%	97.9% ± 0.1%	94.7% ± 0.4%	82.3% ± 2.2%
GCFL+	96.0% ± 0.4%	95.4% ± 0.4%	99.5% ± 0.1%	99.2% ± 0.1%	98.0% ± 0.2%	94.7% ± 0.5%	85.8% ± 1.0%



Heterogeneity level: Mid | family aggregated / pooled performance |
family=flat_balanced | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=flat_balanced | cycle2=25.3%±0.6% | cycle3=26.4%±0.5% | cycle4=26.9%±0.4% | cycle5=26.6%±0.7% | cycle6=26.2%±0.9%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

Best logged test performance (mean ± std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.0% ± 0.1%	95.2% ± 0.1%	99.2% ± 0.1%	98.8% ± 0.2%	97.4% ± 0.2%	94.1% ± 0.2%	86.6% ± 0.4%
FedAvg	96.2% ± 0.1%	96.4% ± 0.1%	99.3% ± 0.0%	98.8% ± 0.2%	97.9% ± 0.1%	95.8% ± 0.3%	90.1% ± 0.3%
FedProx	95.7% ± 0.5%	95.8% ± 0.5%	99.3% ± 0.0%	98.9% ± 0.1%	97.8% ± 0.2%	95.5% ± 0.5%	87.7% ± 1.6%
GCFL+	96.3% ± 0.2%	96.4% ± 0.2%	99.3% ± 0.1%	98.8% ± 0.1%	97.7% ± 0.2%	95.9% ± 0.3%	90.4% ± 0.4%



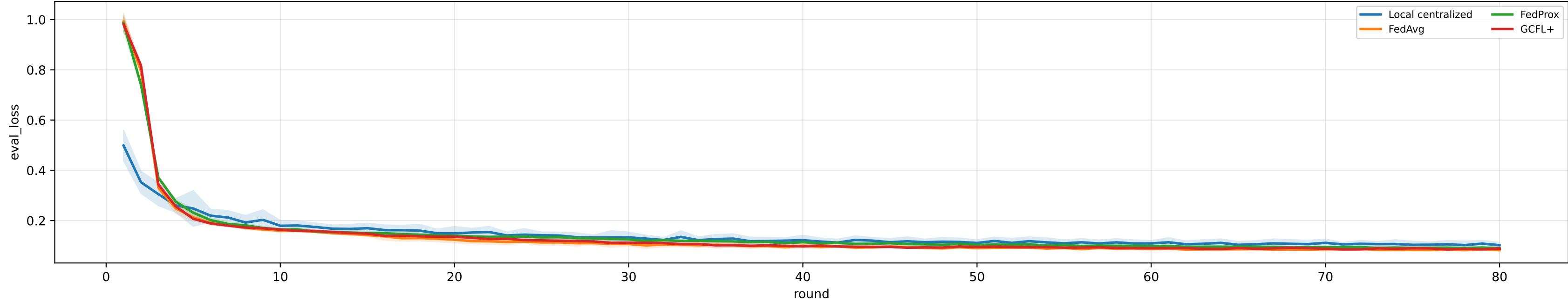
Heterogeneity level: Mid | family aggregated / pooled performance |
family=mild_increasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_increasing | cycle2=18.6%±0.8% | cycle3=22.7%±1.2% | cycle4=25.7%±0.5% | cycle5=27.9%±0.9% | cycle6=28.0%±0.4%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

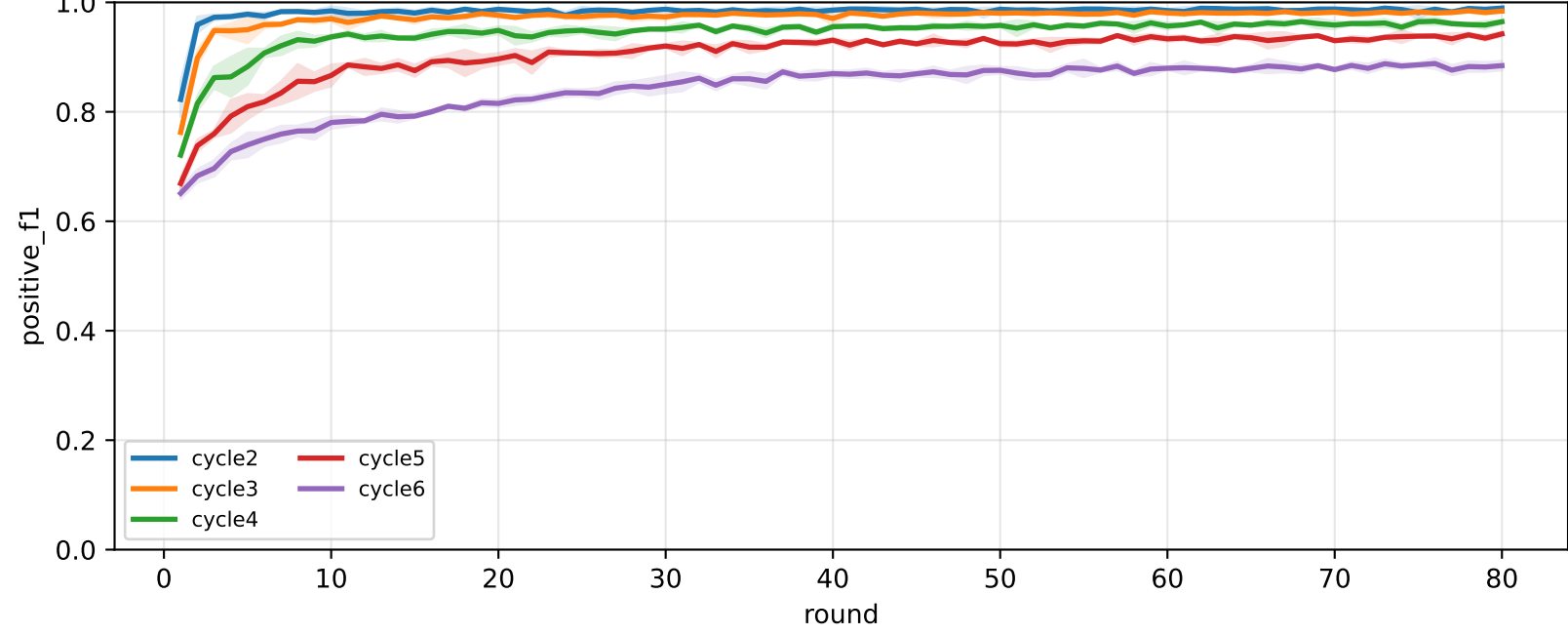
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.9% ± 0.2%	95.4% ± 0.2%	98.9% ± 0.1%	98.2% ± 0.2%	96.8% ± 0.3%	94.1% ± 0.5%	89.2% ± 0.3%
FedAvg	95.9% ± 0.0%	96.3% ± 0.0%	99.0% ± 0.1%	98.3% ± 0.1%	97.5% ± 0.2%	95.1% ± 0.2%	91.6% ± 0.1%
FedProx	95.3% ± 0.5%	95.8% ± 0.4%	99.0% ± 0.1%	98.4% ± 0.1%	97.3% ± 0.2%	94.9% ± 0.5%	89.5% ± 1.4%
GCFL+	95.5% ± 0.3%	95.9% ± 0.3%	98.8% ± 0.2%	98.1% ± 0.1%	97.0% ± 0.2%	94.6% ± 0.6%	91.1% ± 0.5%

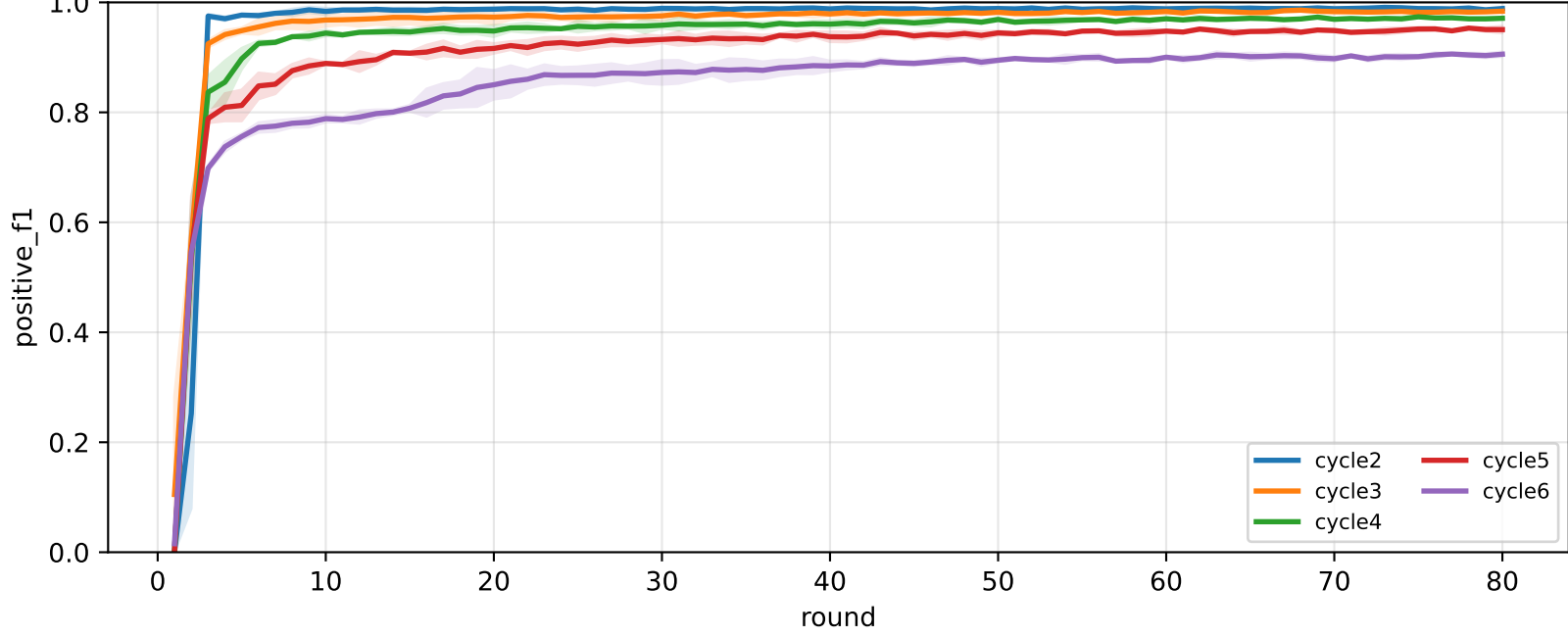
Validation loss comparison



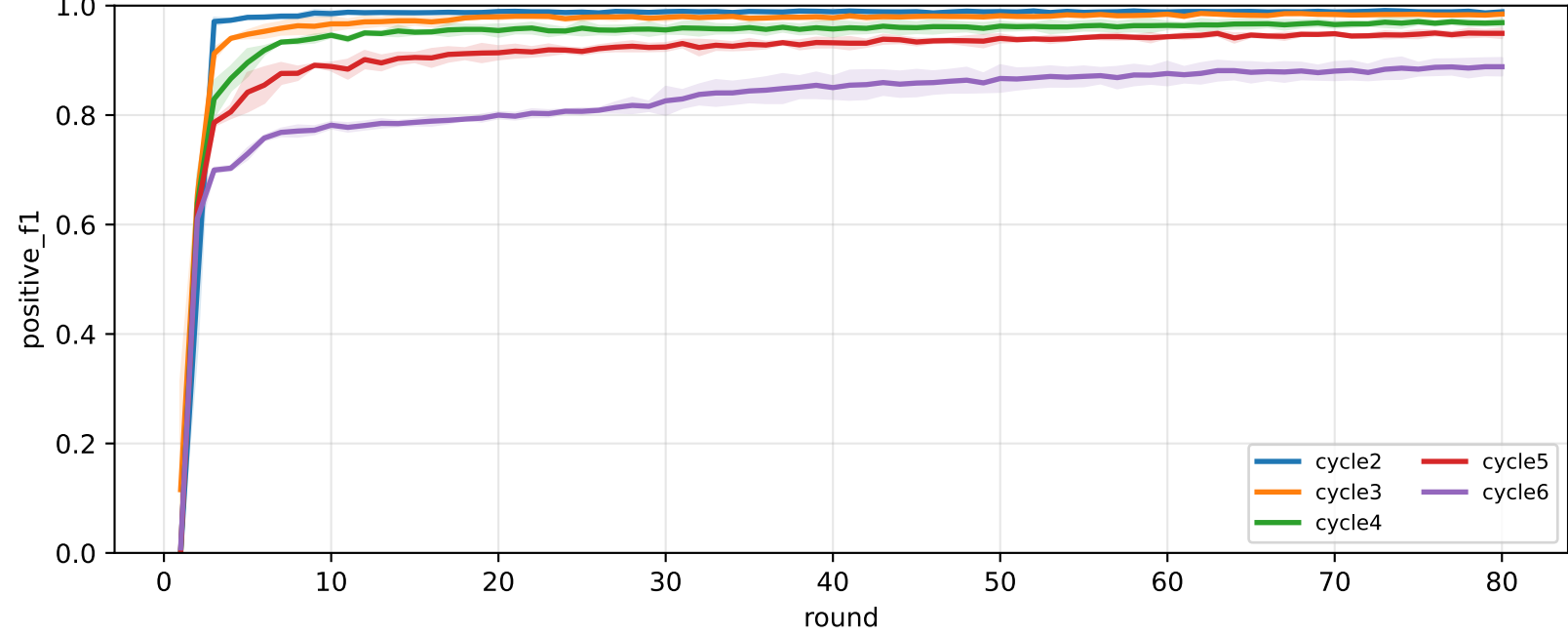
Pooled local centralized positive F1



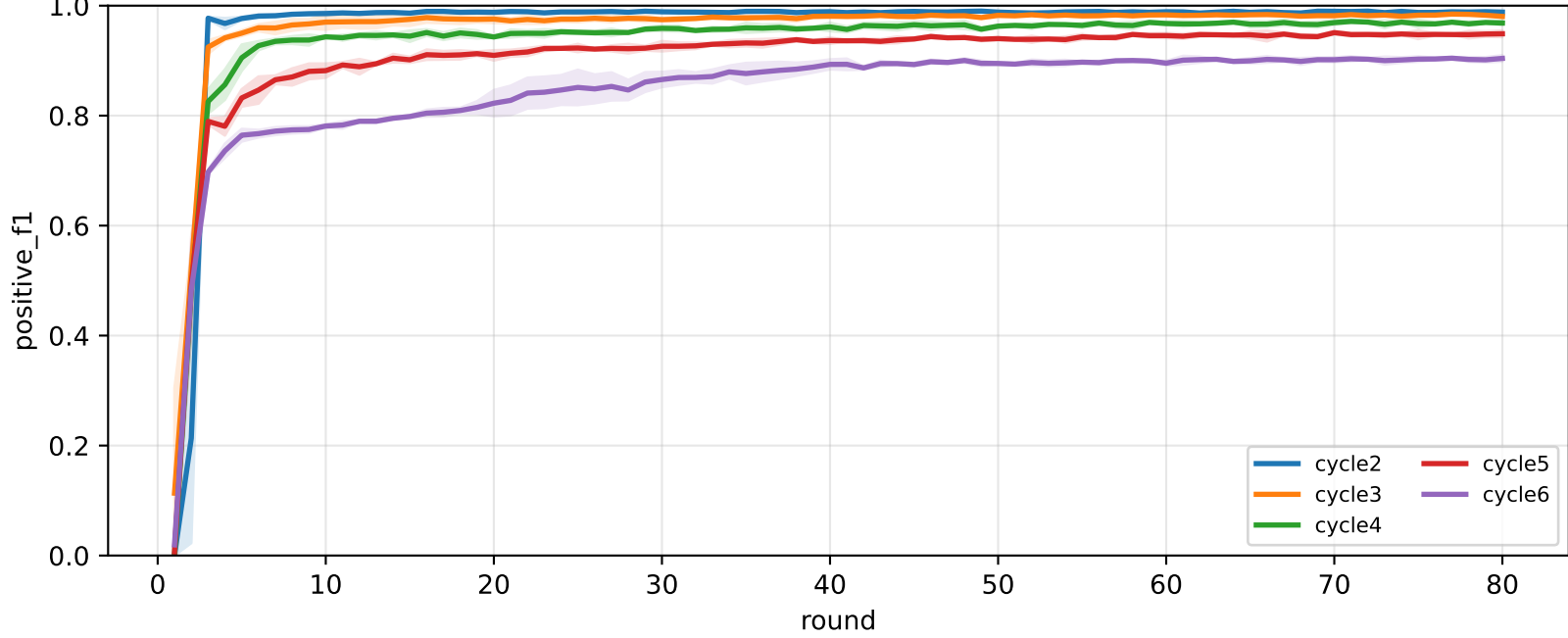
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



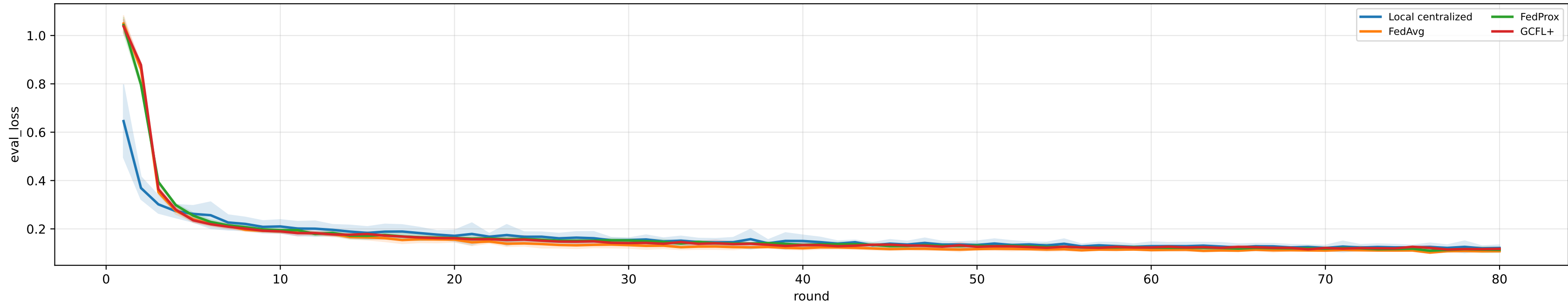
Heterogeneity level: Mid | family aggregated / pooled performance |
family=strong_increasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_increasing | cycle2=10.4%±0.6% | cycle3=15.5%±0.4% | cycle4=21.0%±0.3% | cycle5=24.5%±0.8% | cycle6=28.9%±0.9%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

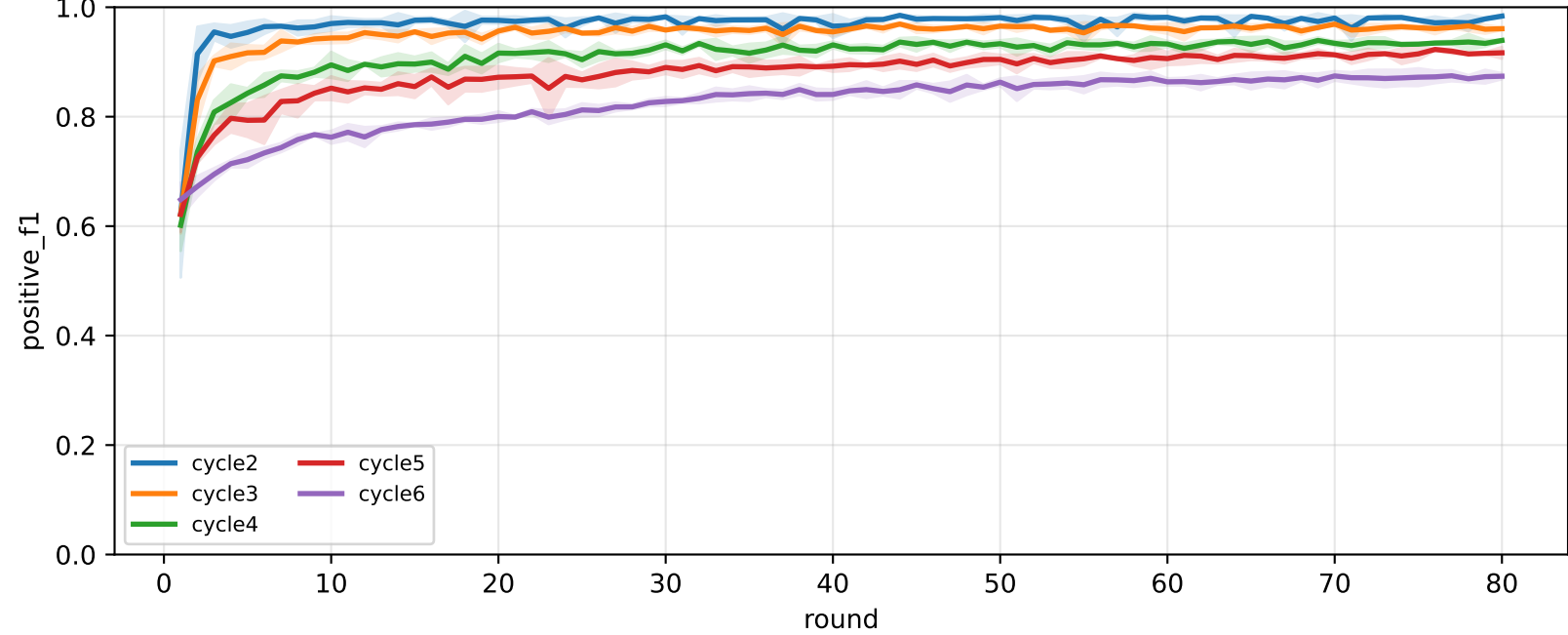
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.1% ± 0.3%	93.4% ± 0.2%	97.9% ± 0.3%	96.5% ± 0.2%	93.9% ± 0.2%	91.1% ± 0.7%	87.5% ± 0.7%
FedAvg	93.7% ± 0.1%	94.7% ± 0.1%	98.3% ± 0.1%	96.8% ± 0.3%	95.4% ± 0.4%	93.1% ± 0.3%	89.9% ± 0.1%
FedProx	92.9% ± 0.7%	94.1% ± 0.5%	98.3% ± 0.1%	97.0% ± 0.2%	94.6% ± 1.2%	92.7% ± 0.6%	88.1% ± 1.1%
GCFL+	92.4% ± 0.3%	93.6% ± 0.2%	98.1% ± 0.3%	96.6% ± 0.1%	94.0% ± 0.3%	91.6% ± 0.5%	87.8% ± 0.8%

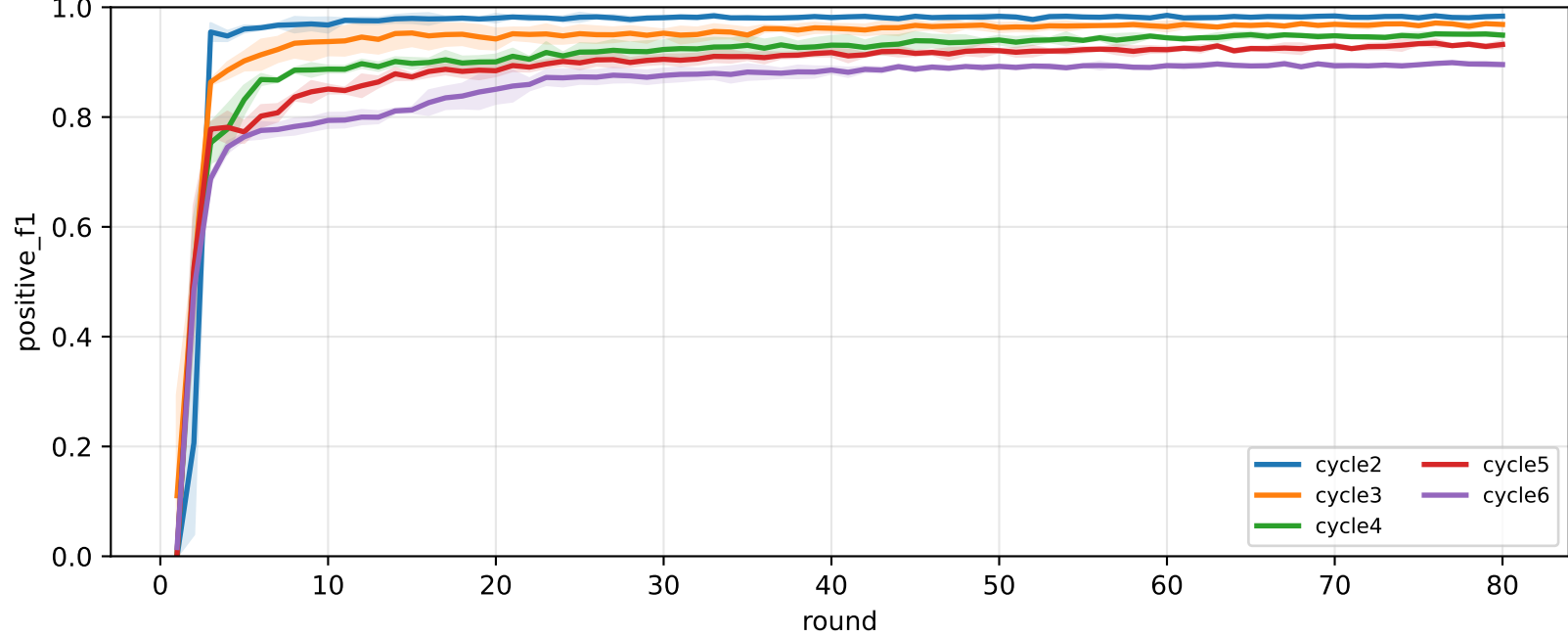
Validation loss comparison



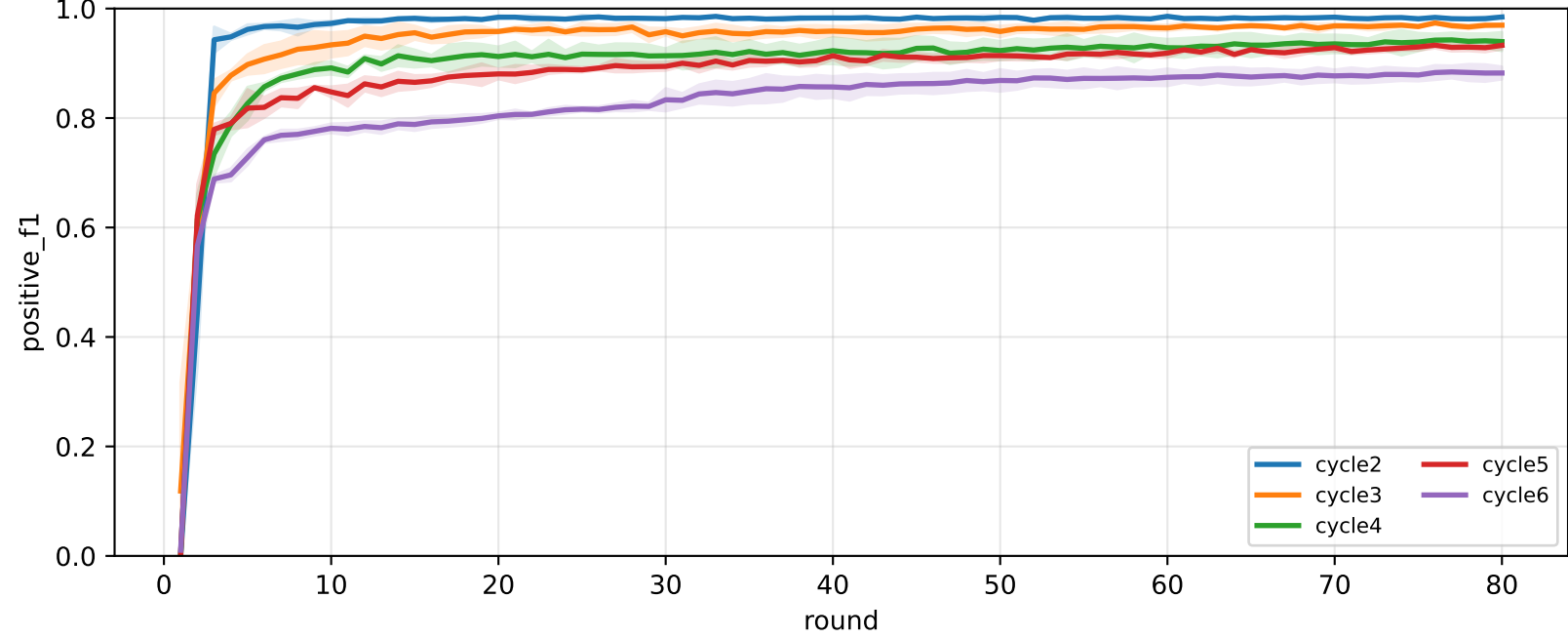
Pooled local centralized positive F1



Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1

